www.tntextbooks.in

The frequency of the oscillation is

$$f=\frac{\omega}{2\pi}=\frac{1}{2\pi}\sqrt{\frac{k}{m}} \text{ Hertz} \qquad (10.23)$$

and the time period of the oscillation is

$$T=\frac{1}{f}=2\pi\sqrt{\frac{m}{k}} \text{ seconds} \qquad (10.24)$$

Notice that ***in simple harmonic motion, the time period of oscillation is independent of amplitude***. This is valid only if the amplitude of oscillation is small.

The solution of the differential equation of a SHM may be written as

$$x(t) = A \sin(\omega t +\varphi) \qquad (10.25)$$

Or

$$x(t) = A \cos(\omega t +\varphi) \qquad (10.26)$$

where A, ω and ϕ are constants. General solution for differential equation 10.21 is $x(t) = A \sin(\omega t +\varphi)+ B \cos(\omega t +\varphi)$ where A and B are contants.

Note

(a) Since, mass is inertial porperty and spring constant is an elastic property.

Time period is $T=2\pi\sqrt{\frac{m}{k}}$

$$T=2\pi\sqrt{\frac{Inertial\ property}{Elasticproperty}}=2\pi\sqrt{\frac{displacement}{acceleration}}$$

(b) $\frac{Displacement}{acceleration}=\frac{x}{\frac{d^2x}{dt^2}}=-\frac{m}{k}$, whose modulus value or magnitude is $\frac{m}{k}$

hence, time period $T=2\pi\sqrt{\frac{m}{k}}$

10.4.2 Vertical oscillations of a spring



Figure 10.14 Springs

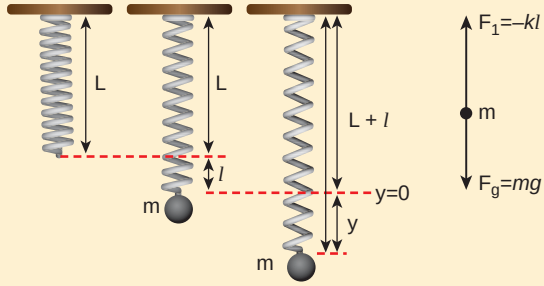


Figure 10.15 A massless spring with stiffness constant k

Let us consider a massless spring with stiffness constant or force constant k attached to a ceiling as shown in Figure 10.15. Let the length of the spring before loading mass m be L. If the block of mass m is attached to the other end of spring, then the spring elongates by a length l. Let F_1 be the restoring force due to stretching of spring. Due to mass m, the gravitational force acts vertically downward. We can draw free-body diagram for this system as shown in Figure 10.15. When the system is under equilibrium,

$$F_1 + mg = 0 \qquad (10.27)$$

But the spring elongates by small displacement l, therefore,